

Relationship between Fuel and Natural Gas Prices over Time

Final Report

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Introduction

This project examines the relationship between fuel and natural gas prices from 2000 to 2023 in Canadian cities. The study focuses on self-distributed gasoline prices for premium and regular octane, as well as natural gas prices for household heating. The objective is to identify the existing relationship between the two commodities and analyze how their prices respond to various factors, including seasonal variations, geographical differences, and time trends. Utilizing a dataset on fuel prices in cents per liter, the project employs a quantitative approach to comparative analysis, providing insightful interpretations while referencing real-world factors such as regional market dynamics and geopolitical events.

The rationale for this project is based on understanding the dynamics of fuel prices over time. It goes beyond simply assessing fuel price trends to examine how two distinct types of fuel—gasoline and natural gas—evolve, and to explore the potential relationship between them. By analyzing their price movements, spreads, and volatility across Canadian cities from 2000 to 2023, the project aims to uncover shared patterns, critical differences, and periods of convergence or divergence that may reflect broader energy market dynamics. Findings from this study will be helpful to numerous stakeholders, including policymakers and government agencies (e.g., Natural Resources Canada), in understanding regional disparities in fuel prices and assessing how seasonal patterns or service models impact consumer costs. It can also support academics and analysts studying market behavior, infrastructure accessibility, and consumer-level fuel pricing dynamics.

Business Impact

This analysis provides valuable insights into the pricing behavior of fuel and natural gas in the energy market. By understanding their relationship and the impact of external factors, shareholders can make more informed decisions regarding energy planning, pricing strategies, and risk management. The findings may also support policymakers in developing targeted and data-driven energy policies and offer researchers a foundation for further studies on market dynamics and price forecasting.

Lastly, the study can enable fuel distributors, logistics companies, and energy firms to optimize pricing strategies, identify underserved markets, and forecast demand more accurately. By uncovering regional and seasonal fuel price patterns, businesses can improve operational planning, reduce supply inefficiencies, and make more informed infrastructure investments.

Data Analysis & Computation

This section provides a detailed explanation of the data source, data wrangling, which involves cleaning, structuring, and addressing missing values and inconsistencies. The exploratory data analysis and statistics analysis discussing the relevant portions of the analytical process, as well as the steps used for analysis, are also included.

- **Dataset**

File name: canada_gas_prices.csv

Description: Monthly panel data containing average gasoline, diesel, and natural gas prices across different Canadian cities from 1979 to 2023

Dataset Details: 42978 rows & 7 columns

Size: 2,431KB (2.37MB)

Source: Kaggle - [Dataset Link](#)

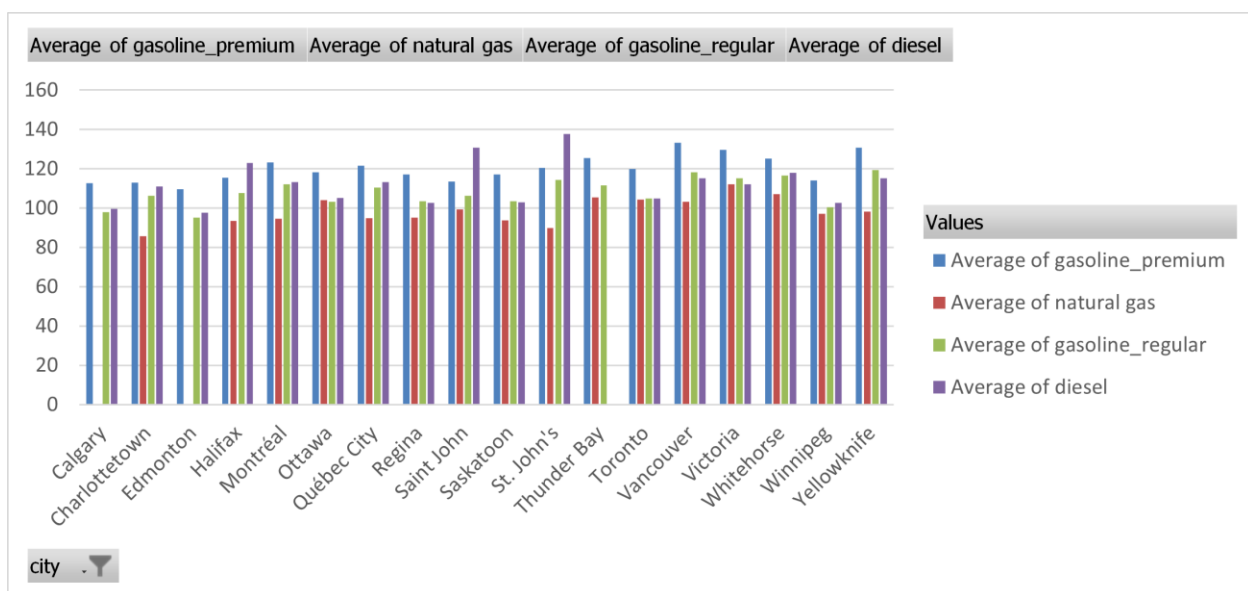
- **Data Wrangling & Cleaning**

Datasets drawn from various sources always require proper preparation and transformation for analysis. The following includes the steps for the data wrangling process:

- **Addressing missing and null values:** The first step taken was to check for any missing or null values in the dataset utilizing the `.isnull().any()` function in pandas. Service and octane columns were identified with null values. As the self-service type covered the larger part of the data, the full-service type was excluded from the sample. The `.fillna()` function was used to replace the remaining null values with a string value “no type” as these were not missing values, but not applicable (household heating had no service, and octane type).
 - **Renaming and Filtering:** To suit the project, household heating was renamed to natural gas. The data was filtered to focus on the years with more concentrated data (2000-2023).
 - **Grouping columns into numerical values:** To represent the fuel types in prices with their respective octane types, the fuel and octane columns were combined to create the fuel type column. Utilizing the `.pivot table()` and `.reset_index()` functions, additional columns with the fuel type and its corresponding octane type in cents per liter were feature engineered.
 - **Sorting of values:** For practical data computation and visualization, the data frame was sorted in ascending order by city, province, and date using the `.sort_values()` function in pandas.
- **Exploratory Data Analysis**

Exploratory data analysis involves a comparative study of the target features in the dataset to achieve the project objective. These include the following:

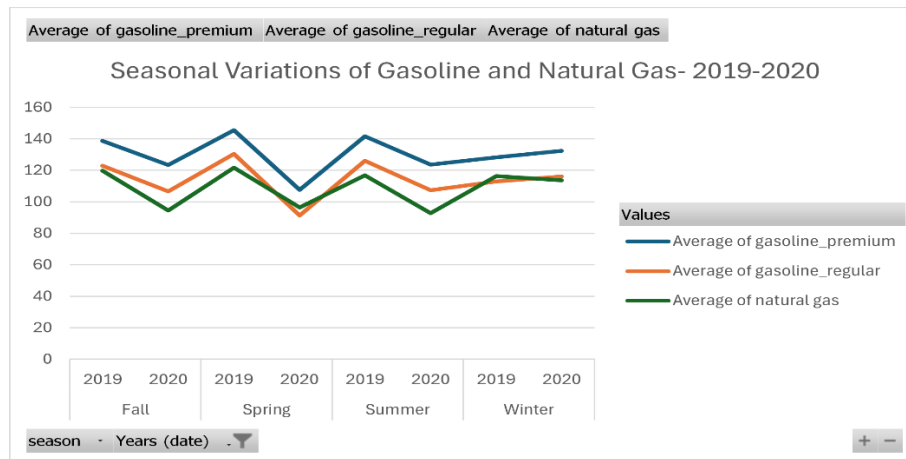
- **Geographical distribution of fuel types:** To establish a representative context for the study, the city and fuel type columns required analysis. A pivot table was used in Excel to group fuel types by their average prices related to their cities. Column charts were used to create visualizations for analysis and data presentation. The figure below shows the average prices of fuel types by city.



It is worth noting that 15 cities have uniform data on all fuel types, while Calgary and Edmonton have no records of natural gas, and Thunder Bay has no data on diesel. Therefore, diesel is excluded from the sample, resulting in an increase in the number of cities to 16.

- **Seasonal Variations:** To measure the seasonal variation in fuel prices, the date column was converted to a datetime format to create the year and month columns, leveraging the `to_datetime` and `dt` accessor functions in Pandas. The seasonal column was then feature engineered using the `map()` function to group the months into winter, spring, summer, and fall. A line chart was used for the first analysis of the seasonal patterns. The line chart below

illustrates the seasonal variations in fuel type from 2019 to 2020. consistent seasonal drop in prices during Spring 2020 for all fuel types, likely

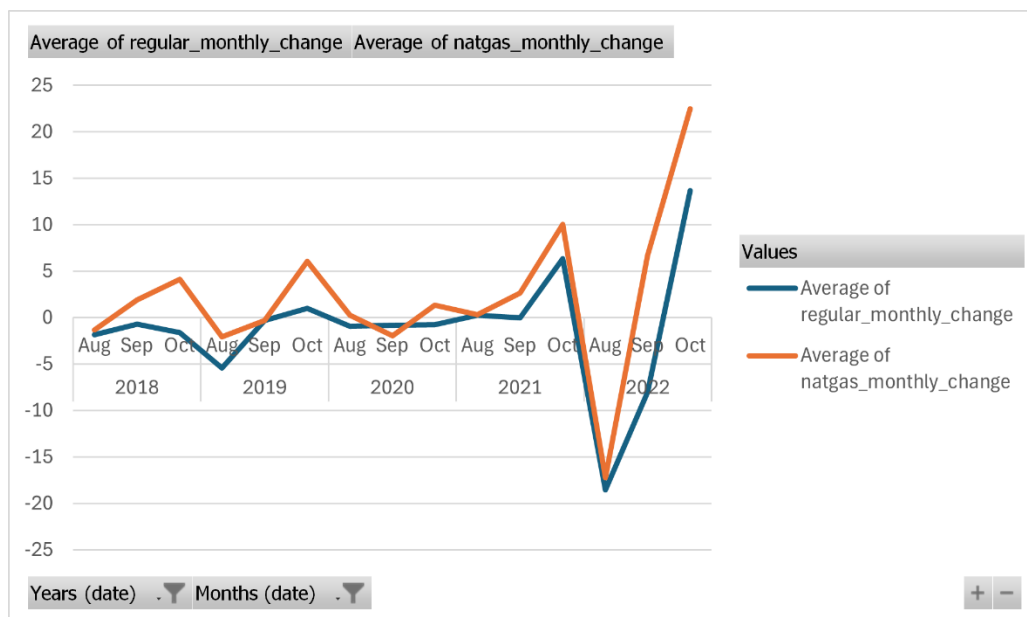


- **Computing the price spread of fuel types:**

To conduct this analysis, I calculated the monthly price difference between premium gasoline and natural gas for each city in the dataset using the ``diff()`` function in pandas. Specifically, for every month of the year, the average price of natural gas was subtracted from the average price of premium gasoline. These differences were then analyzed across the full-time span to identify consistent patterns or notable anomalies in the price relationship. The results indicated that premium gasoline has typically been more expensive than natural gas across most cities and years. However, a notable exception occurred in 2022, during which gasoline prices were lower than natural gas prices in several months — particularly in fall and winter.

- **Analyzing fuel and natural gas price volatility:** Price volatility offers important insight into the stability of fuel markets and helps identify periods of significant disruption or unusual price behavior. To conduct this analysis, I computed the monthly price differences for each fuel type using a time-series approach while leveraging the ``diff()`` function in pandas. For every city and province in the dataset, the price for each month was compared to the

previous month, producing a series of price change values that reflect the degree and direction of movement over time. This allowed for the identification of fluctuation patterns and the comparison of fuel price stability between gasoline and natural gas. The line chart below illustrates the volatility of regular gasoline and natural gas prices over 5 years from August to October. It clearly shows that while the magnitude of change



varied, the direction of price movement between the two fuels was often aligned. In many periods, both gasoline and natural gas either increased or decreased in the same months. This pattern suggests that both fuel types are likely influenced by shared macroeconomic forces, such as inflationary pressure, global oil and gas supply dynamics, or geopolitical disruptions.

- **Computing the standard deviation:** Basic statistics were performed in pandas to compute the standard deviation of the monthly changes in the fuel prices to assess the degree of volatility associated with each fuel type. This statistical measure captures the average dispersion of monthly price changes from the mean, providing a quantifiable indicator of how stable or erratic the prices are over time. For example, in 2001 the average standard deviation of premium gasoline was calculated to be 6.75 cents per liter, while natural gas

showed a slightly lower standard deviation of 6.5 cents per liter. These findings support the visual analysis presented in the line chart: although natural gas experienced some large price shifts in certain months (particularly in 2022), regular gasoline displayed more frequent smaller fluctuations, which cumulatively contributed to a slightly higher overall volatility.

Challenges and Solutions

The dataset had several limitations which restricted comparative analysis for the study. Not all cities had complete data for all fuel types (e.g., some lacked natural gas or diesel prices in certain months or years). This limited the ability to make uniform comparisons across all locations and time periods. The data did not include whether fuel prices were after tax or before tax. Also, data on full service were incomplete and had to be excluded from the sample.

Missing values were carefully handled during data cleaning. The study focused the analysis only on cities and time frames with reliable records to ensure meaningful results.

Dashboard Description

The dashboard ([Dashboard Link](#)) is designed to analyze and visualize the relationship between premium and gasoline prices compared to natural gas prices in Canadian cities from 2000 to 2023. It highlights key indicators such as price volatility over time, seasonal fluctuations, geographical differences of fuel types, price differences and standard deviation of monthly changes in price. The primary use case is to support energy analysts, researchers, and policymakers in understanding fuel pricing behavior over time and geography. Users can interact with filters to explore different time frames and price fluctuations, view price patterns by cities, and switch between dashboard views focused on premium or regular gasoline.

Conclusion

This study analyzed fuel price behavior across 16 Canadian cities using 4,432 records spanning from 2000 to 2023. The goal was to understand the relationship between gasoline (both regular and premium) and natural gas, focusing on volatility, seasonal variation, and price gaps.

The results indicate that gasoline prices are generally more volatile than natural gas. While natural gas experienced larger price jumps during specific periods, especially around late 2022, gasoline displayed more frequent month-to-month changes, contributing to a slightly higher overall volatility.

Seasonal variation further highlighted that certain seasons—particularly winter and fall—tend to exhibit sharper fluctuations in fuel prices, suggesting that weather conditions, heating demand, and supply logistics may influence pricing differently across fuel types. In 2022, an exceptional pattern emerged when gasoline prices temporarily dropped below natural gas prices during the colder months in almost all cities—an inversion likely driven by the global energy crisis, which disproportionately affected natural gas supply and pricing.

Lastly, while gasoline and natural gas prices occasionally move in the same direction, the overall correlation is not consistent, implying that the two fuels are influenced by shared macroeconomic and energy market trends, but still behave independently due to their distinct use cases, supply chains, and seasonal demand cycles.

Future Works

While this project provided valuable insights into the relationship between gasoline and natural gas prices across Canadian cities, there are several directions for future research that could further enhance our understanding of fuel market dynamics.

One key area for future work is the integration of fuel demand data. By incorporating information on consumption patterns—such as monthly or seasonal fuel usage across cities or provinces—researchers could explore how demand levels influence price volatility and price relationships. This would add an important economic dimension to the analysis, allowing for the assessment of how consumer behavior interacts with market trends.

Additionally, analyzing demand data would help determine whether gasoline and natural gas function as substitutes or complementary goods. For example, if an increase in gasoline prices leads to a rise in natural gas consumption, it could suggest substitution. Conversely, if demand for both fuel types moves together, it may point toward complementarity, possibly linked to broader energy consumption patterns or multi-fuel systems (e.g., dual-fuel heating or transportation).

Future projects could also benefit from the inclusion of other influential variables, such as government policy changes, carbon pricing, fuel tax regulations, or supply disruptions, to more comprehensively assess the external factors that shape fuel price behavior. Leveraging predictive modeling techniques like regression analysis or machine learning could further enhance forecasting capabilities and support policy evaluation.

